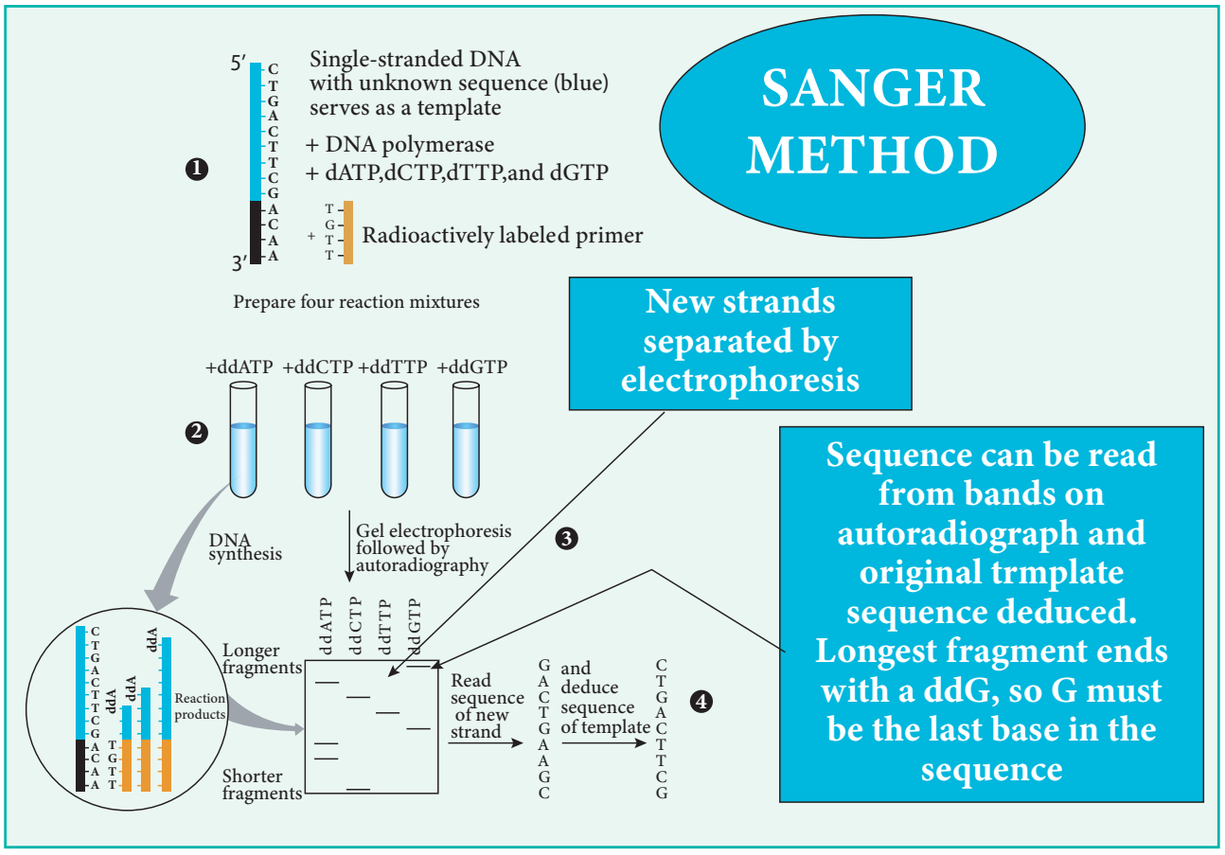


Figure 19 Schematic representation of Sanger sequencing procedure

6.6.2 The Next Generation Sequencing (NGS) Technologies and their role in genomics.

All the Next Generation Sequencing (NGS) methods adopt the principle of Sanger's method i.e., sequencing-by-synthesis. However they all use technologies that do not permenantly terminate the growing chain, so that the synthesis can be monitored as each base is incorporated. This improvement allows us to conduct sequencing of millions of DNA fragments simultaneously and hence achieving high throughput, which was not possible in Sanger's method. Moreover the sequencing costs has become much lower using these technologies and today a human genome can be sequenced at around $200 and within 48 hours. Some of the popular methods are the Illumina reversible dye terminator method, ion torrent method and nano pore DNA sequencing methods.

The advancements in genome sequencing has led to more ambitious projects like the 1000 genomes project or 1KGP. It was an international collaboratory effort to sequence 1000 different human genomes and to catalogue their genetic variations. This project was started in 2008 and was completed in 2012 with the complete genome sequence of 1098 humans. With the help of this data, so far about 100 regions in human genome has been identified to have association with diseases like coronary heart disease, diabetes mellitus etc.,

The Cancer Genome Atlas program (TCGA) is another ambitious project which was started in 2007 and so far has collected genomics data from 20,000 cancer patients and includes 33 types of cancer. This data is freely available for researchers around the world

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